Assignment One

Data Modeling using ER Diagram and Relational Data Model

Canada Flower Store

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EECS 3421: Introduction to Database Systems

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1. **Understand the business functions**

Canada Flowers is an online flower store recognized as an award-winning Canadian Florist. The company supports same-day flower deliveries across Canada and also facilitates international flower orders for a wide variety of occasions and sentiments. In addition to flower deliveries, they offer flower subscriptions and gift deliveries services.

The website they use has several features to support its business functions. It is organized into multiple pages—*Occasions, Sentiments, Gifts* and *International*—each containing categories that allow customers to browse products easily (see Figure 1).

On the *Occasions* page, flowers are organized into categories such as Anniversary, Anytime, Birthday, Budget, Maternity, Funeral, Easter, Mother's Day, Father's Day, Spring, Summer and Fall flowers. Each category offers bouquets created for the occasion. For instance, the Fall flowers section highlights arrangements with orange and yellow tones, in contrast, the Funeral Flowers section emphasizes white floral arrangements. The *Sentiments* page organizes flowers into categories such as Congratulations, Friendship, Get Well, Love & Romance Flowers, Sympathy and Thank You. The *Flowers* page groups bouquets by New Arrivals, Best Sellers, Flower Varieties, Colour and more. The *Gifts* page features gift cards, food and wine, fruit and gourmet baskets, indoor plants, and flower subscriptions. Additionally, the website has a search function which enables users to quickly find specific products based on their preferences (such as Figure 2).

When a user clicks on a bouquet (see Figure 3), they are directed to the product's page, which displays a description, flower types, available product version, as well as the product name and ID. Customers can then add the product to their cart for purchase with optional add-ons.

At checkout, users may proceed as guests or sign in/create an account. If a user decided to make an account, the create account page (see Figure 4) collects basic information including full name, email address, street address, and phone number.

Finally, the checkout process is completed in several steps. First, users enter delivery details (see Figure 5), including delivery address, date, type, and recipient information such as full name, address, phone number, and gift information (card message and signature). Next, users complete the purchase by providing payment and billing information, which includes full name, email, phone number, an optional alternate phone, address, and card details. This page may differ slightly depending on whether the user is logged in or checking out as a guest (see Figure 6).

2. **State your business rules and assumptions**

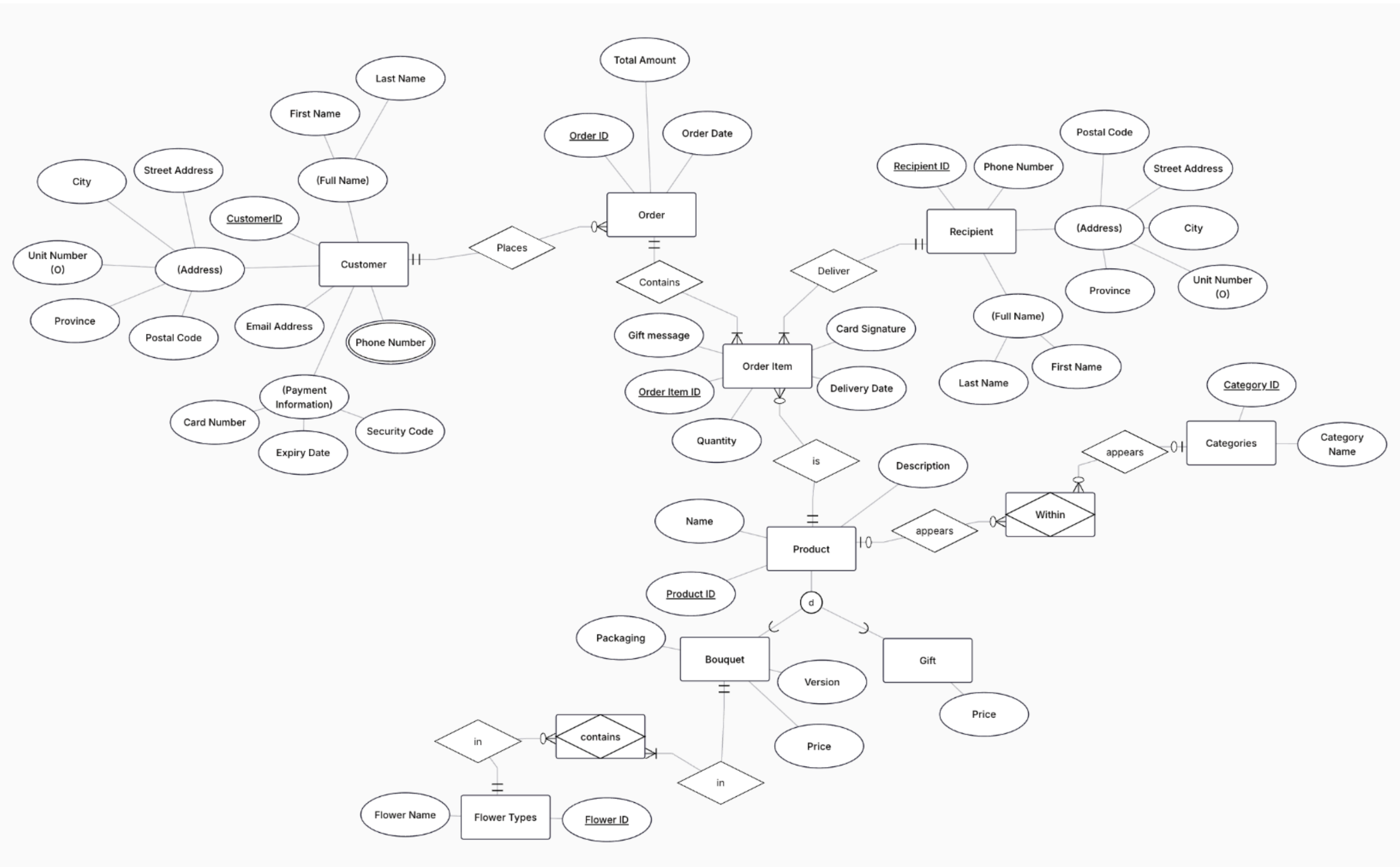
For this project, I am unable to fully replicate the database style used by the Canada Flowers website. However, to replicate it as closely as possible, I will make the following business assumptions, simplifying some of the original business functions:

- A customer must use a valid credit card to make an online payment.
    - For this assignment, I will be treating registered and guest users equally.
- An order can contain multiple products and each order has a unique Order ID

- Each product in an order can be delivered to anywhere in North America, but only to a single location per product. Each product will also have a specialized gift message and card signature.
- Delivery dates for the purchase must be for the same day or a future date.
- Each product (flowers or gifts) must belong to at least one category (e.g. Occasions, Sentiment, etc)
- Products may appear in multiple categories (for example, a Rose Bouquet can be featured under *Flower*: Rose **and** *Sentiment*: Love & Romance)
- Each product has a unique ID and name
- Each registered account will contain a single email address, address, full name, and may have up to 2 phone numbers.

International orders, optional product add-ons, and flower subscriptions will be excluded from this database. Furthermore, I have also disregarded the gift categories (gift cards, food & wine, fruit & gourmet) listed on the website.

3. **Build an E-R model**
   The entire E-R model at a glance is depicted below.



In the E-R Model, there are nine entities in total, including two subclasses. Below, I outline each entity with its key functions and provide explanations for attributes that require clarification.

- Customer: Information about the customer (individual that places the order).
    - Address is a composite attribute, which contains an optional unit number, for individuals that live in apartments.
    - Phone number is a multivalued attribute as individuals may enter up to two phone numbers.
    - Full name is a composite attribute that collects first name and last name.
    - Payment information is a composite attribute that collects card number, expiration date and the security code for the credit card.
- Order: Contains general order information.
- Order Item: Information about specific products in an order.
- Product: Masterlist of all products available.
    - Bouquet: Information about each of the flower arrangements.
    - Gift: Information about each of the gift products.
- Flower Type: Contains information about all the flower types.
- Recipient: Collects information about the recipients of the purchases.
    - Address is a composite attribute, which contains an optional unit number, for individuals that live in apartments.
    - Full name is a composite attribute that collects first name and last name.
- Categories: Contains information about all the categories on the website, occasions and sentiments.

Product is divided into two subclasses, Bouquet and Gift. since the subclasses share common attributes; however, Bouquet also contains distinct attributes that Gifts does not possess. For this model, Gifts are considered all gift cards, indoor plants, or food & wine delivery boxes (as listed on the website). Bouquet possesses the packaging attribute which indicates whether the bouquet will be wrapped or delivered in a vase. The version attribute determines which bouquet variety will be delivered– standard, upgraded or premium. The Bouquet subclass also contains a relationship that the Gift class does not possess. This relationship will be described below.

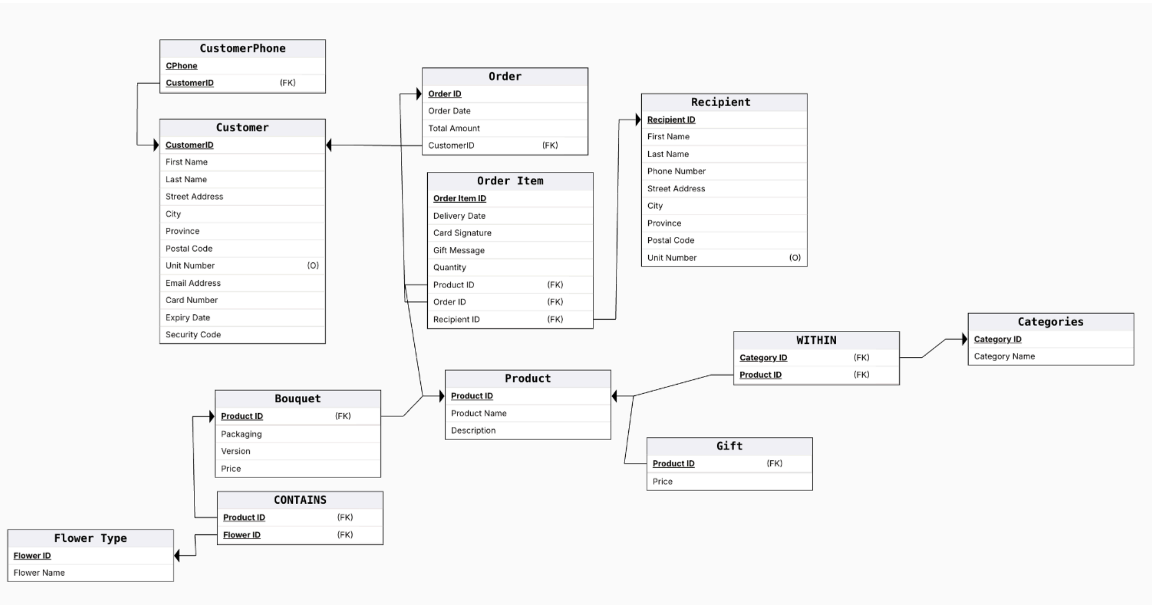
Existing relationships:

- There exists a relationship, *places*, between Customer and Order. Customers can place (not mandatory) many orders, but each order must belong to one customer.
- There exists a relationship, *contains*, between Order and Order Item. Each order must contain at least one order item, and may contain many order items. Each order item must belong to only one order.
- There exists a relationship, *deliver*, between Order Item and Recipient. Each order item must be delivered to one recipient. A recipient may be sent/delivered several order items.
- There exists a relationship, *is*, between Order Item and Product. Many products can be considered an order item (not mandatory - in the case the product in not ordered). Each order item must be considered as a single product on the website.
- There exists a relationship between Bouquet and Flower Types. Since this relationship is a many to many relationship, I have created an associative entity, *contains*. Each

bouquet can contain several flower types, and must contain at least one flower type. In contrast, a flower type can show up in many bouquets. The flower type relationship is optional as a flower type can exist in the website without being in a bouquet, the website may just sell the flower type on its own.

- There exists a relationship between Categories and Product; however, this relationship is a many to many relationship, so I have created an associative entity, *Within*. Each product many belong to several categories, but a product <u>does not have to</u> belong to a category. Each category can have several products, but it does not have to have a product associated with it to be displayed on the website (could add the product at a later time).

4. Convert your E-R model to relational tables



The next section will include images of the tables with sample data.

5. Normalization check

<u>Customer Phone Table</u>

- There are no functional dependencies, the only columns are primary keys
- Table is in 3NF - atomic, no partial keys, no transitive dependencies
- No normalization is required.
- This table assumes that customer phone is a multivalued attribute - customers may have up to two phone numbers

Customer Phone

CPhone	Customer ID
416-555-5019	C1
416-555-5020	C1
416-591-5555	C2
647-970-5555	C3
647-416-9274	C3

Customer

Full Name / Address / Payment Info.											
	Full Name		Address						Payment Info.		
Customer ID	First Name	Last Name	Street Address	City	Province	Postal Code	Unit (o) Number	Email Address	Card Number	Expiry Date	Security Code
C1	Steve	Jobs	123 Apple St.	Toronto	Ontario	A1B 2C3		Steve @gmail.com	—	—	—
C2	Tate	McRae	127 Apple St.	Toronto	Ontario	A1B 2C7		tate @gmail.com	—	—	—
C3	Sabrina	Carpenter	155 Blue Ave.	Toronto	Ontario	A1C 9S4	102	Sabrina @gmail.com	—	—	—
C4	Dave	Jobs	143 Cake Hill	Toronto	Ontario	C2G 4T6		Dave @gmail.com	—	—	—
C5	Kim	Sam	155 Blue Ave.	Toronto	Ontario	A1C 9S4	324	Kim @gmail.com	—	—	—
C6	Holly	Jay	721 Rose St.	Toronto	Ontario	B2J 5L1	401	Holly @gmail.com	—	—	—

Assume that these cells are filled with card information

<u>Customer Table</u>

- Functional dependencies:
    - Customer ID → Full Name (First/Last Name)
    - Customer ID → Address (Street Address/City/Province/Postal Code/Unit Number)
    - Customer ID → Email Address
    - Customer ID → Payment Information (Card Number, Security Code, Expiry Date)
- All are full key dependencies and the table is in 3NF since all of the cells are atomic and there are no existing partial or transitive dependencies.
- No normalization of the table is required.

<u>Order Table</u>

- Functional Dependencies:
    - Order ID → Order Date
    - Order ID → Total Amount
    - Order ID → Customer ID
        - Assuming each order is placed by one customer.
- All are full key dependencies and the table is in 3NF since all of the cells are atomic and there are no existing partial or transitive dependencies.
- No normalization of the table is required.

Order

Order ID	Order Date	Total Amount	Customer ID (FK)
0111	July 1, 2025	$710.50	C1
0112	July 2, 2025	$116.50	C2
0113	July 2, 2025	$99.00	C5
0114	July 10, 2025	$51.00	C4
0115	Aug 1, 2025	$451.97	C1

Recipient

	Full Name			Address				
Recipient ID	First Name	Last Name	Phone Number	Street Address	City	Province	Postal Code	Unit (o) Number
R1	Dave	Jobs	647-970-1111	143 Cake Hill	Toronto	Ontario	C2G 4T6	
R2	Holly	Jay	416-555-5134	721 Rose St.	Toronto	Ontario	B2J 5L1	401
R3	Kim	Sam	647-416-~~7777~~	155 Blue Ave.	Toronto	Ontario	A1C 9S4	324
R4	Tate	McRae	416-591-5555	127 Apple St.	Toronto	Ontario	A1B 2C7	
R5	Steve	Jobs	416-555-5019	123 Apple St.	Toronto	Ontario	A1B 2C3	

<u>Recipient Table</u>

- Functional dependencies:
    - Recipient ID → Full Name (First/Last Name)
    - Recipient ID → Address (Street Address/City/Province/Postal Code/Unit Number)
    - Recipient ID → Phone Number
- All are full key dependencies and the table is in 3NF since all of the cells are atomic and there are no existing partial or transitive dependencies.
- No normalization of the table is required.

Order Item

Order Item ID	Delivery Date	Card Signature	Gift message	Quantity	Product ID (FK)	Order ID (FK)	Recipient ID (FK)
F123	July 1, 2025	—	—	1	BD57FA	0111	R2
F124	July 2, 2025	—	—	1	BD26FA	0111	R1
F125	July 3, 2025	—	—	1	SB29AA	0111	R3
F227	July 2, 2025	—	—	1	BD57FA	0112	R2
F228	July 3, 2025	—	—	1	BD110AA	0112	R2

*Assume Card Signature + Gift messages are filled with messages

<u>Order Item Table</u>

- Functional dependencies:
    - Order Item ID → Delivery Date
    - Order Item ID → Card Signature
    - Order Item ID → Gift Message
    - Order Item ID → Quantity
    - Order Item ID → Product ID
        - Assuming each order item IS a product
    - Order Item ID → Order ID
        - Assuming each order item BELONGS TO an order
    - Order Item ID → Recipient ID
        - Assuming the order item will be DELIVERED to one recipient
- All are full key dependencies and the table is in 3NF since all of the cells are atomic and there are no existing partial or transitive dependencies.
- No normalization of the table is required.

<u>Gift Table</u>

- Functional dependencies:
    - Product ID → Price
- All are full key dependencies and the table is in 3NF since all of the cells are atomic and there are no existing partial or transitive dependencies.
- No normalization of the table is required.

Gift

Product ID	Price
FAW124AA	$121.00
PL1FA	$75.00
PL8FA	$108.00
PL46AA	$75.00
FG3TA	$130.00

Product

ProductID	Product Name	Description
SB29AA	Blissful Season	—
BD57FA	Sweet & Pretty	—
AN46TA	Blush Life	—
FG16AA	The Gala	—
FAW92AA	Holiday Hug	—

Product Table

- Functional dependencies:
  - Product ID → Product Name
  - Product ID → Description
- All are full key dependencies and the table is in 3NF since all of the cells are atomic and there are no existing partial or transitive dependencies.
- No normalization of the table is required.

Bouquet

Product ID	Packaging	Version	Price
AN46TA	vase	Standard	$84.00
AN46TA	vase	upgraded	$101.00
AN46TA	vase	premium	$120.00
BQ21AA	wrapped	Standard	$35.00
BQ21AA	wrapped	premium	$60.00

Bouquet Table

- Functional dependencies:
  - Product ID → Packaging
  - Product ID → Version
  - Product ID → Price
- All are full key dependencies and the table is in 3NF since all of the cells are atomic and there are no existing partial or transitive dependencies.
- No normalization of the table is required.

WITHIN

Product ID (FK)	Category ID (FK)
BD56TA	3
BD44AA	3
AV13FA	2
AV13FA	3
AV5TA	1

Within Table (Associative Entity)

- There are no functional dependencies, the only columns are primary keys
- Table is in 3NF - atomic, no partial keys, no transitive dependencies
- No normalization is required.

Categories Table

- Functional dependencies:
  - Category ID → Category Name
- All are full key dependencies and the table is in 3NF since all of the cells are atomic and there are no existing partial or transitive dependencies.
- No normalization of the table is required.

Categories

Category ID	Category Name
1	Anniversary
2	Anytime
3	Birthday
4	ThankYou
5	Get Well

CONTAINS

Product ID (FK)	Flower ID (FK)
AN48TA	2
AN48TA	3
AN58TA	1
AN58TA	3
AN58TA	5

<u>Contains Table (Associative Entity)</u>

- There are no functional dependencies, the only columns are primary keys
- Table is in 3NF - atomic, no partial keys, no transitive dependencies
- No normalization is required.

<u>Flower Type</u>

- Functional dependencies:
    - Flower ID → Flower Name
- All are full key dependencies and the table is in 3NF since all of the cells are atomic and there are no existing partial or transitive dependencies.
- No normalization of the table is required.

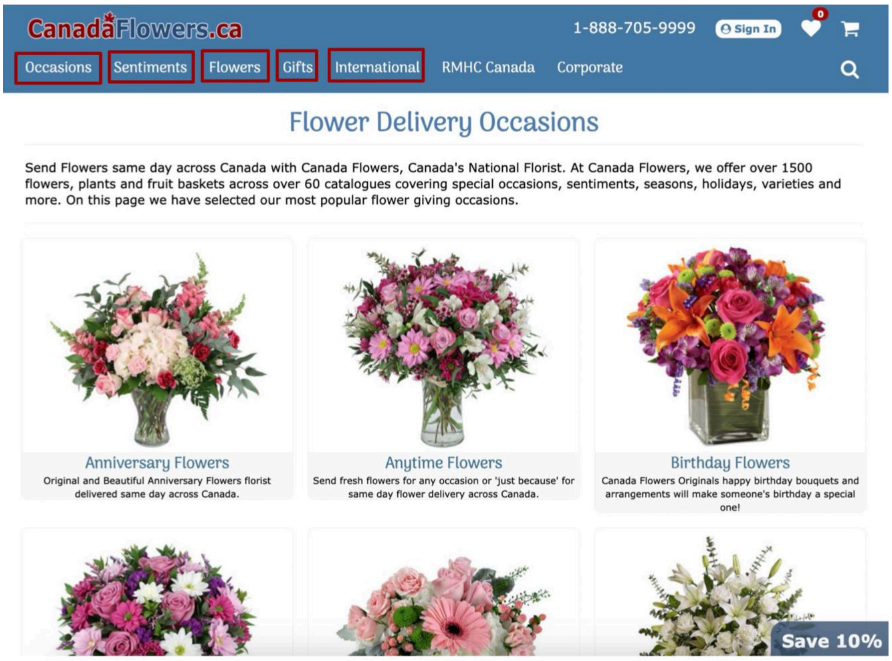
Flower Type

Flower ID	Flower Name
1	Rose
2	Carnation
3	Statice
4	Asiatic Lily
5	Greenery

Appendix

Figure 1: Web Page

"Occasions" webpage - searching for flower delivery for specific occasions



Similar webpages are found for the other highlighted pages

Figure 2: Search Result

Searching flower assortments with roses

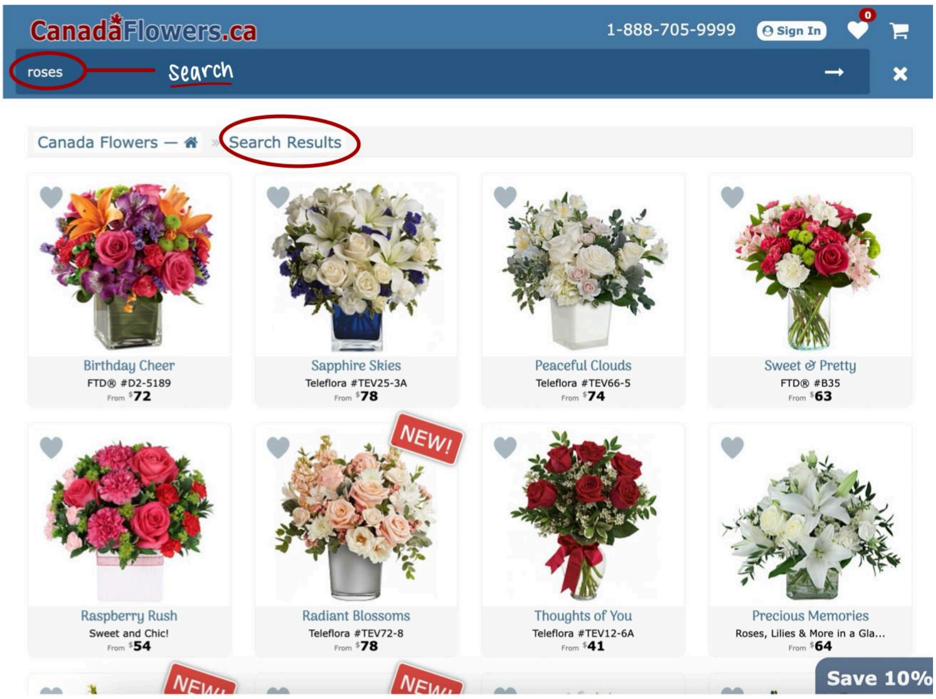


Figure 3: Product Page

Product details and descriptions for Teleflora's Radiant Blossoms Bouquet

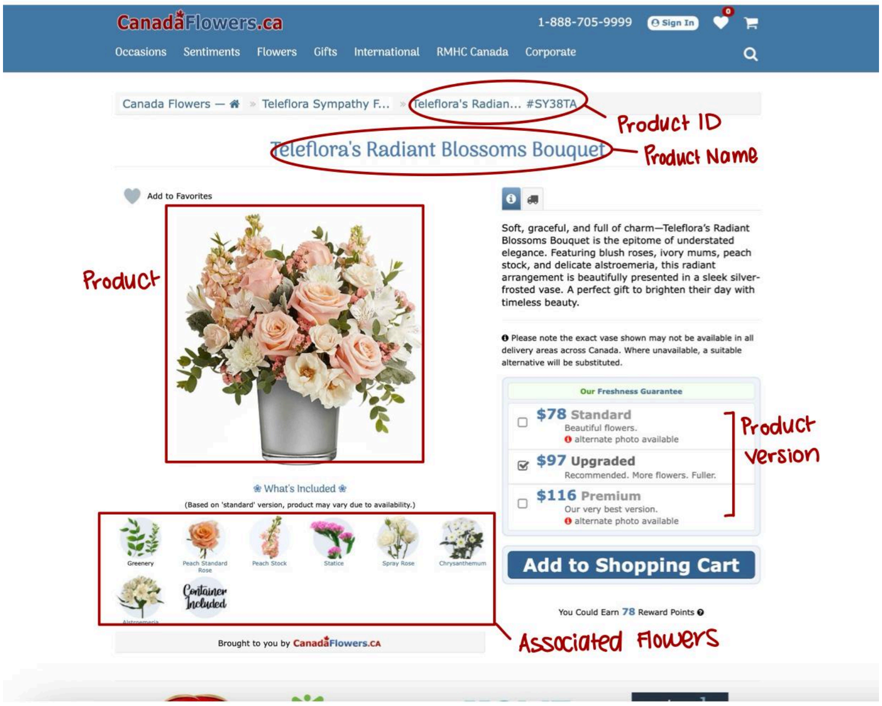


Figure 4: Creating Account Page

Filling in personal information to create an account.

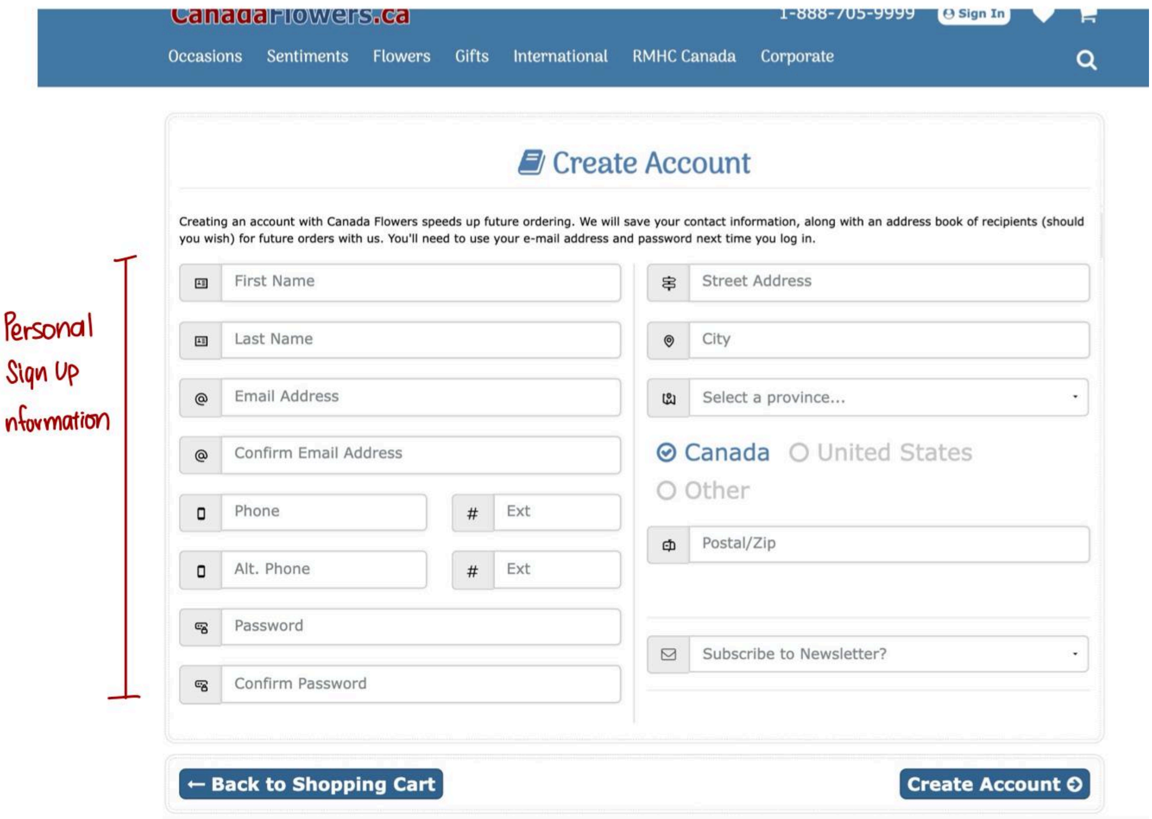


Figure 5: (Guest) Cart Page

Filling in recipient information and delivery information for (each) product/purchase.

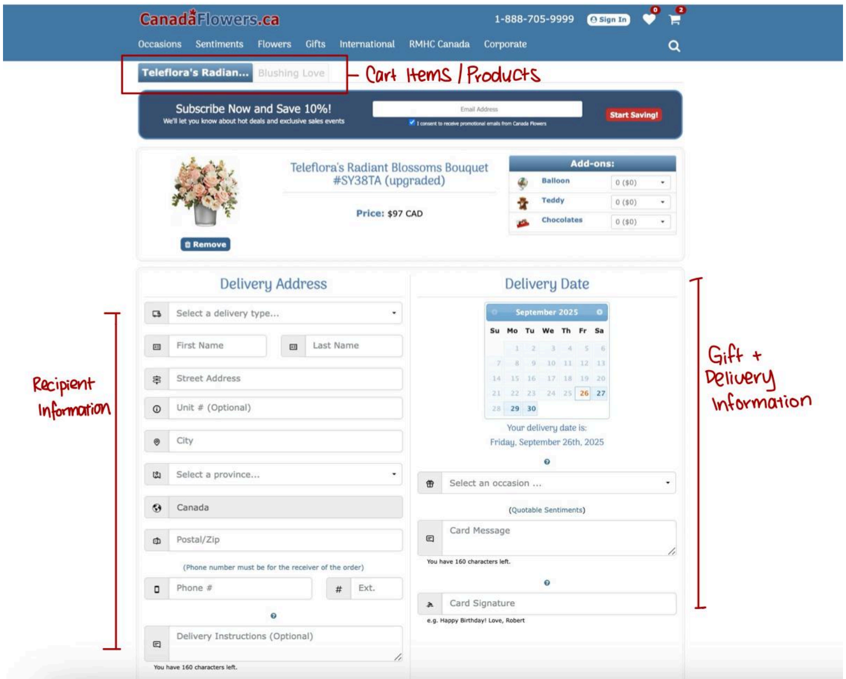


Figure 6: (Guest) Cart Page 2

Filling in payment information for purchase.